

Reflection Journal Report for Lab 03

Prepared for Prof. Vishwantha Rao, HCC ITAI 1371 Machine Learning

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The purpose of this report is for each student to reflect on their learning journey for ITAI 1371 Machine Learning (ML) Lab 03. This Reflection Journal Report consists of separate parts for each student in the group. Each student in the group was responsible for their own content and report formatting in each of the sections below.

This reflection journal entry is for Lab 03 Google Collab Gradients and was prepared by Irene "Alex" Ramos.

Academic Reflective Journal for Irene Ramos Lab 3

1. Introduction:

The current reflective journal looks into the completion of a basic Machine Learning (ML) lab related to classification algorithms, ML process, and feature selection. The aim of the reflection is to examine my practical experience of developing a Logistic Regression algorithm with the use of a dataset for classifying wines, theoretical implications of data handling, and application of these concepts in the field of cybersecurity.

2. Description of Experience or Topic:

The lab experiment helped gain hands-on experience in the three main areas of machine learning: supervised learning, unsupervised learning, and reinforcement learning. The main exercise was to build a logistic regression model that would be able to classify three different classes of wine according to their chemical makeup. An important part of this exercise included trying to use feature selection and determining which features would work best to improve upon the baseline model.

3. Personal Reflection:

Initially, it appeared paradoxical how reducing the amount of data inputted into the model could increase the accuracy of the algorithm. However, since the model was able to identify the most relevant predictors, it is easy to see why the presence of “noisy” data could throw off an algorithm. This supports the theory that for a machine learning model based on the distance calculations, there must be clear demarcation points between classes. A review of the work process reveals that although the selected subset of features contributed to increasing the accuracy of the model, there were still issues with the scalability of the features. Given that while some features, like proline, occur in the hundreds, other features, like flavonoids, occur in the single digits, using a scaling algorithm, such as StandardScaler, would be the mathematically appropriate next step.

4. Discussion of Improvements and Learning:

This assignment helped me understand the data pre-processing stage even better and realize just how important it is for the algorithm's performance. I have learned how to work with scikit-learn, how to evaluate models and how to recognize the features that can predict certain outcomes. Apart from the current data set, there are many other applications for this knowledge including information security. Unsupervised anomaly detection to recognize zero-day attacks on a network will require the same feature selection process.

5. Conclusion:

In conclusion, this lab has shown me that machine learning goes far beyond just using an algorithm; it comes down to truly knowing your data. Through this lab, I have learned that selecting features is just as important as selecting the correct algorithm, changing my perception from simply coding to crafting the data itself. In the future, regardless of whether it is academic modules or cyber security applications, I know that these concepts will continue to be at the forefront of my mind.

This reflection journal entry is for Lab 03 Google Collab and was prepared by Eric Roberson.

I. Understanding Insights & New Breakthroughs

I already knew about the types of learning from my Intro to AI course. The three types of learning are as follows:

- Supervised
- Unsupervised
- Reinforcement

I've created the following table to compare the three:

TYPES OF MACHINE LEARNING

TYPE:	Supervised	Unsupervised	Reinforcement
DEFINITION:	Learns from labeled input-output pairs to predict outputs for new data	Finds hidden structure or groupings in data that has no labels	Learns by trial and error, choosing actions that maximize reward
DATA USED:	Labeled (input paired with correct output)	Unlabeled (no predefined outputs)	No fixed dataset; an agent interacts with an environment
FEEDBACK SIGNALS:	Correct answer is known and directly compared	None — evaluation is more subjective	Delayed rewards or penalties from the environment
GOALS:	Map inputs to correct outputs as accurately as possible	Discover patterns, clusters, or structure on its own	Maximize cumulative reward over a sequence of decisions
APPLICATIONS:	Spam detection, disease diagnosis, price prediction	Customer segmentation, anomaly detection, topic modeling	Robotics, game playing, autonomous vehicles

II. Analyzing the Problem / Problem Solving Strategies

I did not face any challenges with the lab, per se. From my reading, I have a better understanding of overfitting, which is when you are using supervised learning and you have too much training data labeled, and ML cannot learn and apply knowledge to the test sample. This was a new concept for me, and I can see how working with very large amounts of data could become a problem. In my current role, I work with some applications that have 40TB MS SQL databases. I'm not sure that ML would be wise to use against that much data, at least not in a production environment with live production data. Also, there are databases that are bigger than 40 TBs, so I'm curious how ML works with real-world data. I'm also curious if you can use a big data lake (blob) as test data. My inkling is that you can, but it is probably beyond the scope of what we will do in labs.

In Part 7 Hands-on Practice portion of the lab, I used the following features (variables) and these were the accuracy of the results for each:

```
Your model features: ['alcohol', 'malic_acid', 'ash']  
Your model accuracy: 0.806 (80.6%)  
Original model accuracy: 0.889
```

```
Your model features: ['alcohol', 'ash', 'magnesium']  
Your model accuracy: 0.694 (69.4%)  
Original model accuracy: 0.889
```

```
Your model features: ['alcohol', 'magnesium', 'flavanoids']  
Your model accuracy: 0.944 (94.4%)  
Original model accuracy: 0.889
```

Here's the conclusions I can draw about comparing the data:

- The alcohol / magnesium / flavanoids feature combination was my best-performing model at 94.4%, actually beating the original all-features model (88.9%).
- The alcohol / ash / magnesium combination was my worst at 69.4%.
- My takeaway is that more features is not automatically better – what matters is whether the features you pick actually separate the three wine classes.
- Flavanoids appears to be a strong discriminator between cultivars, so pairing it with alcohol gave the model a clear signal to work with, while ash and magnesium don't vary as sharply between classes and added more noise than signal.

If I wanted to improve further, I'd try scaling the features with StandardScaler (since alcohol and proline are on very different numeric ranges), and I'd look at a correlation heatmap against the target class before picking features rather than guessing. Everything I did in the lab with features was a random selection — essentially a guess. Obviously, there would be a plan constructed with real-world data rather than just randomly making feature selections as I did with this lab. So, being intimately familiar with your data is super important with real-world data.

III. Real-World Applications

Since I work in IT and since the department I am in is smashed in the middle of the technology stack, I work with lots of databases. In enterprise database administration, and from the perspective of my role as a Resiliency Lead, I'm curious about how ML can help prevent disasters. In using ML, it would seem it could be used to predict some production database performance or capacity issues (e.g. flagging a database heading toward disk, memory head room, or

query-latency thresholds) before they become incidents. It seems that supervised learning would easily apply to existing databases because they are arrays of data and essentially already labeled. Since my role is risk and controls, one way to use ML would be to predict capacity based on historical metrics (CPU, I/O, query times, storage growth). Then, label whether an incident followed a certain performance benchmark, which would train a classifier or regression model to predict risk. In fact, I believe our Capacity Analytics software from BMC is doing just that since they have integrated artificial intelligence into BMC's Capacity Analytics software. I work with all kinds of databases (Oracle, MS SQL, Cockroach, and PostgreSQL to name a few), and our Capacity Analytics software definitely monitors disk space, memory, etc regardless of the database type, operating system (even mainframes). I am not certain it is using ML to predict the likelihood of future events—rather it's just monitoring real-time performance. ML can be used to make predict future outcomes. Regardless of the database or OS manufacturer, the same data would be needed, which would include the following:

- time-series performance metrics from monitoring tools,
- historical incident/ticket records,
- database or disk size and growth trends,
- query performance logs
- external data sources such as KBs or release notes

This ties back to the question I raised above about how ML handles very large datasets (40TB+ SQL databases) – in this use case you wouldn't train against the live production data itself, but against aggregated performance metrics collected over time, which keeps the training set manageable. Would it be feasible for a company to replicate a 40 TG set of data to have test data that is not live in production? To me, the cost would be huge. Obviously, other areas in which ML could be applied would be in the area of finance. It would be the same as studying the bottlenecks for databases but instead to predict market conditions based on historical events from the past. Although, I think financial markets are much more volatile than the databases and OSes I support!

IV. Connecting New & Prior Knowledge

My prior knowledge only consisted of the types of learning—supervised, unsupervised, reinforcement—not the situations in which they can be applied. I think the most basic is supervised learning where data has to be labeled (tagged). I think the most advanced type of learning is reinforcement, and I am basing this on how that particular type of learning is applied (e.g. autonomous vehicles, robotics). I could be mistaken, but this is the connection I have made

from this lab. Please refer to the table above to see how I've connected new knowledge.

V. Emerging Questions

I have the following emerging questions after Lab 03:

1. How much data is too much? Can ML handle 40 TBs of data or a blob or Hadoop?
2. How does neural networks tie into ML? (Architecture)
3. What about compute power? Obviously, you'll need more compute power with real world data. But how much?
4. Where does weights and back propagation come in?
5. There has to be more to overfitting. Will we dive into that more later?
6. How much does it cost to implement ML? If you are not using live production data for training or testing, and it is lots of data, would the cost be exorbitant?

VI. Changing Approach & Critical Thinking

I am still early in my learning about AI and very early in the semester for this course, so I do not believe I have the requisite knowledge to change my approach or critical thinking about ML yet. I still feel like I'm missing something vital in this course like training material, or a book, or some type of learning videos. I will say that it is absolutely necessary to process the .IPYNB file on Colab to understand the material. Regrettably, I am considering dropping this course and come back to it once I have more knowledge from other classes.

This reflection journal entry is for Lab 03 Google Collab Gradients and was prepared by Xiomara Torres:

Introduction:

In this journal i will examine my experience completing the module 3 lab. This lab introduces three main types of machine learning. The activity demonstrated in this lab focuses on using chemical measurements from different wines to predict their wine class.

Description of Experience or Topic:

The lab began by explaining supervised, unsupervised, and reinforcement learning. I learned that supervised learning uses labeled data, meaning that the correct answers are provided during training. Unsupervised learning uses data without known answers and reinforcement learning involves an agent learning through trial and error by receiving rewards or punishments.

This specific portion of the lab focused on supervised learning.

Personal Reflection:

In the beginning I had a hard time understanding what I was looking at or what exactly I was supposed to learn from running the cells. It started to make more sense to me when i used the output as the answer key the model used to learn. I learned how training and testing data must be separated. Testing the model on information it has already seen would not show whether it can make accurate predictions on new data. I was surprised that logistic regression performed better than the Decision Tree because i assumed the Decision Tree would be more advanced. This showed me that a model is not the better choice because it's more complicated, which is why different models need to be tested and compared.

Discussion of Improvements and Learning:

This lab showed me that the features given to a model can affect how well it performs. When three different features were used, the model's accuracy decreased, showing that fewer features do not automatically produce better results. In the future i would test more features with different numerical ranges

Conclusion:

Overall, i learned that machine learning involves more than running an algorithm, the data must be understood and tested using information the model has not already seen. although

i still need to practice interpretating code and evaluating results, this lab gave me a better understanding of how the different stages of machine learning work together.

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